

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

SUBMMITED TO:

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Lab Assignment 4: Selection Sort

Experiment No. 4.1 - Index of the smallest value

Experiment Title:

Index of the smallest value

Aim:

To find the index of the smallest value in an array.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Selection sort divides the array into a sorted part and an unsorted part. In each step it finds the smallest value of the unsorted part and swaps it into the first unsorted position. It makes at most one swap per pass, but always needs $O(n^2)$ comparisons.

Algorithm / Procedure:

1. Create the array {64, 25, 12, 22, 11} and set minIndex = 0.
2. Loop from index 1 to the end.
3. If a[i] < a[minIndex], set minIndex = i.
4. Print minIndex.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 4, Task 1
// Question: Find the index of the smallest value in the array {64,
// 25, 12, 22, 11} with a loop, and print that index.

public class Main {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};
        int minIndex = 0;

        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[minIndex]) {
                minIndex = i;
            }
        }

        System.out.println(minIndex);
    }
}
```

Input:

Fixed array declared in the program: {64, 25, 12, 22, 11}.

Output:

```
4
```

Time / Space Complexity: Time O(n); Space O(1).

Result:

The program was compiled and executed successfully. The smallest value 11 was located at index 4.

Viva-Voce Questions:

- Why do we store the index instead of the value?
- What is the role of minIndex?
- What is the time complexity of finding the minimum?

Experiment No. 4.2 - Selection sort

Experiment Title:

Selection sort

Aim:

To sort an array using the selection sort algorithm.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Selection sort divides the array into a sorted part and an unsorted part. In each step it finds the smallest value of the unsorted part and swaps it into the first unsorted position. It makes at most one swap per pass, but always needs $O(n^2)$ comparisons.

Algorithm / Procedure:

1. Create the array {64, 25, 12, 22, 11}.
2. For each position i, assume minIndex = i.
3. Scan positions i + 1 to the end and update minIndex when a smaller value is found.
4. Swap a[i] with a[minIndex].
5. Print every value of the sorted array.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 4, Task 2
// Question: Sort the array {64, 25, 12, 22, 11} with selection sort.
// For each position, swap in the smallest value of the rest. Print
// each value of the sorted array on its own line.

public class Main {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};

        for (int i = 0; i < a.length - 1; i++) {
            int minIndex = i;

            for (int j = i + 1; j < a.length; j++) {
                if (a[j] < a[minIndex]) {
                    minIndex = j;
                }
            }

            int temp = a[i];
            a[i] = a[minIndex];
            a[minIndex] = temp;
        }
```

```
        for (int i = 0; i < a.length; i++) {
            System.out.println(a[i]);
        }
    }
}
```

Input:

Fixed array declared in the program: {64, 25, 12, 22, 11}.

Output:

```
11
12
22
25
64
```

Time / Space Complexity: Time $O(n^2)$ in all cases; Space O(1).

Result:

The program was compiled and executed successfully. The array was sorted in ascending order: 11, 12, 22, 25, 64.

Viva-Voce Questions:

- What is the time complexity of selection sort?
- How many swaps does selection sort make per pass?
- How does selection sort differ from bubble sort?

Experiment No. 4.3 - First step of selection sort

Experiment Title:

First step of selection sort

Aim:

To perform only the first step of selection sort.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

Selection sort divides the array into a sorted part and an unsorted part. In each step it finds the smallest value of the unsorted part and swaps it into the first unsorted position. It makes at most one swap per pass, but always needs $O(n^2)$ comparisons.

Algorithm / Procedure:

1. Create the array and set minIndex = 0.
2. Find the index of the smallest value with a loop.
3. Swap a[0] with a[minIndex].
4. Print every value of the array.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 4, Task 3
// Question: Do only the first step of selection sort on {64, 25, 12,
// 22, 11}: find the smallest value, swap it with the first value,
// and print each value of the array on its own line.

public class Main {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};

        int minIndex = 0;

        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[minIndex]) {
                minIndex = i;
            }
        }

        int temp = a[0];
        a[0] = a[minIndex];
        a[minIndex] = temp;

        for (int i = 0; i < a.length; i++) {
            System.out.println(a[i]);
        }
```

```
    }
}
```

Input:

Fixed array declared in the program: {64, 25, 12, 22, 11}.

Output:

```
11
25
12
22
64
```

Time / Space Complexity: Time O(n); Space O(1).

Result:

The program was compiled and executed successfully. After the first step the array is 11, 25, 12, 22, 64.

Viva-Voce Questions:

- After the first step, which part of the array is sorted?
- Why is the swap needed?
- How many such steps are needed for n elements?